

CSCE 580 Quiz 1

1. About data for AI [4 + 16 = 20 points] Instructions: Give your answers in bullet points.
  - a. What is open data? Given an example of open data that you produce which others can use? [2 + 2 = 4]
    - i. Open data is data that has been made publicly available and is free to use for other purposes. It is often published by authorities, such as a country or city, and is updated regularly. The data may vary in quality, but it is important that the data is organized and documented properly.
    - ii. A potential open dataset I could produce and allow others to use may be weather-related data. I could track measures such as temperature, humidity, wind speed, etc along with the data and location and make the data accessible to others.
  - b. You are analyzing a dataset and some attributes are missing.
  - c. What could be any 2 reasons why they are missing? [2 + 2 = 4]
    - i. Human error could be responsible for the missing values. A person might have mistakenly forgot to put that value into the dataset.
    - ii. Depending on the type of data, the value may have been purposefully omitted. A participant in a survey or a medical patient may have opted to keep the information private.
  - d. What are any 2 ways you can still proceed with data analysis despite the missing values. For each, mention what assumption you are making and what are its risks. [(2+2+2) \* 2 = 12]
    - i. You could choose to leave out observations that are missing an attribute that you need. This might make more sense if very few observations have missing values. This method assumes that removing the entire entry will not have a significant impact on the results. The risk is it could have a noticeable impact on the results of your analysis, especially if there might be an underlying reasons why the data was omitted i.e. a distinct group is more likely to be omitted.
    - ii. Another method is to perform data imputation. This involves creating value to fill in for the missing data, often by averaging or another method. This method assumes that the imputed value is somewhat close to the true value. Imputation using the mean assumes that the data is normally distributed. A potential risk is the imputed value may be very different from what the true value is, potentially throwing off your analysis.